

CHAPTER 9

PSYCHOLOGICAL FOUNDATIONS OF PERFORMANCE

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After completing this chapter, you will be able to

- understand the psychological constructs of arousal, motivation, focus, and confidence and be able to ascertain their impact on physical performance;
- comprehend terms relevant to psychological areas of concern, such as anxiety, attention, and the ideal performance state;
- understand signs and symptoms of mental health issues, predisposing factors to injury, psychological recovery, techniques to handle fear of reinjury, introduction of rehabilitation-type strength and conditioning, performance psychology concepts, arousal management concepts such as self-efficacy, imagery, goal setting, progressive muscle relaxation, and centering;
- understand varying ways to manipulate practice schedules including coaching psychology of motor learning, whole–part,

- random, and variable practice and how to use these schedules to facilitate skill acquisition and learning; and
- understand different types of instructions and feedback and their application in a practice and performance setting.

Excellence in athletic performance is the result of sound skill and physical training accompanied by optimal rest and recovery cycles and appropriate diet. The role of sport psychology is to help athletes achieve more consistent levels of performance at or near their physical potential by carefully managing their physical resources through appropriate psychological strategies and techniques. By understanding these strategies and techniques, strength and conditioning professionals can better design sport-specific and even position-specific training programs with the ultimate goal of maximizing performance.

After introducing foundational concepts, we address how one's mind, through cognition, can influence physical performance, and then we describe the **ideal performance state**—the ultimate goal of every athlete. In part, this state is marked by psychological and **physiological efficiency** (i.e., employing only the amount of mental and physical energy required to perform the task). We discuss the primary psychological influences—motivation, attention, and arousal—on skill acquisition and performance, citing several theories of how these phenomena can change psychomotor learning and athletic performance. Understanding mental health and the impact of injuries on psychological states of athletes is discussed. Finally, we describe coaching techniques and cues, including goal setting, energy management and relaxation, imagery, and confidence development, that can be employed to enhance overall performance in strength and conditioning environments, as well as other performance venues.

Role of Sport Psychology

An athlete is someone who engages in a social comparison (i.e., competition) involving psychomotor skill or physical prowess (or both) in an institutionalized setting, typically under public scrutiny or evaluation. The essence of athletic competition involves comparing oneself to others and putting ego and self-esteem on the line in a setting that is bound by rules and regulations. The psychologically well-prepared athlete is characterized by efficiency of thought and behavior. Efficiency is typically associated with skilled performance, when actions are fluid and graceful, which conserves energy. The concept can also be extended to psychological activity; an efficient athlete adopts a task-relevant focus, not wasting attention on task-irrelevant processing such as worrying, catastrophizing, or thinking about other things such as critical external observers such as a coach or audience.

Sport psychology is a multifaceted discipline, drawing on constructs of exercise science and psychological principles, that seeks to understand the influence of behavioral processes and cognitions on sport performance. Sport psychology is typically classified as a scientific field of study within sports medicine and has three major goals:

- Measuring psychological phenomena
- Investigating the relationships between psychological variables and performance
- Applying theoretical knowledge to improve athletic performance

By applying the information gained through an awareness of sport psychology principles, athletes can better manage their physical resources, thereby producing more effective performances. In reality, many athletes come into the training and competitive environment with some rudimentary mental skills, but there is little understanding of how these skills evolved or even how to best use

them for effective performance. Throughout this chapter we introduce a structure for understanding the interrelationships not only between the mental skills themselves but between the mental skills and the physical, technical, and tactical skills being developed in the weight room and on the practice field. Note, however, that these skills can only be truly effective if they are understood, practiced, and applied in a performance setting. Like the physical, technical, and tactical constructs described throughout the rest of this book, mental skills also need to be taught, practiced, integrated into performance, and evaluated for effectiveness.

Ideal Performance State

The ideal performance state has been studied from a number of perspectives. Williams and Krane (68) listed the following characteristics that athletes typically report about this state:

- Absence of fear—no fear of failure
- No thinking about or analysis of performance (related to the motor stage of automaticity)
- A narrow focus of attention concentrated on the activity itself
- A sense of effortlessness—an involuntary experience
- A sense of personal control
- A distortion of time and space, in which time seems to slow

In a sense, this ideal performance state seems to represent everything that applied sport psychology programs attempt to promote. There is an absence of negative self-talk, a strong feeling of efficacy, and an adaptive focus on task-relevant cues. An important aspect is that the athletes trust in their skill and conditioning levels and just let it happen, without interference from negative cognitive processes.

Kobe Bryant, a premier player in the National Basketball Association, described being in this state, saying,

When that happens, you really do not try to focus on what's going on because ... you could lose it in a second. Everything becomes one noise—you don't hear this or that; everything's just one noise—you're not paying attention to one or the other.... You just really try to stay in the present and not let anything break that rhythm. Again, as long as you just kind of stay there, you become oblivious to everything that's going on. You don't think about your surroundings or what's going on with the crowd or the team. You're kind of locked in.... You have to really try to stay in the present and not let anything break that rhythm.

Bryant's comments richly reinforce many of the concepts discussed throughout this chapter. It is important to remember that his mental state rested largely on a sound physical training program and on a history of performance success. Bryant exhibited phenomenal physical prowess, arduously running sprints, training on the court, and lifting weights in the off-season. Combined with superior performance on the basketball court, such preparatory physical effort contributed greatly to his focused, confident psychological state.

Energy Management: Arousal, Anxiety, and Stress

In order for athletes to perform effectively, they need to learn how to best manage their mental and physical energy levels. Athletes who waste energy through worry, anger, frustration, or anxiety experience a greater likelihood of distraction and decreased self-confidence, and they have less physical energy for when performance is most important (20). Thus, the ability to maintain self-control and manage energy reserves in a performance environment is a critical skill for any performer.

Mental energy is generated, maintained, depleted, and refreshed via our emotions. **Emotions** are temporary feeling states that occur in response to events and that have both physiological and psychological components (19). These emotions affect mental and physical energy and therefore can have both beneficial and detrimental effects on human performance. Emotions such as excitement can be beneficial to performance, causing us to feel motivated, which elevates self-confidence and reinforces commitment levels. However, emotions can be detrimental when an athlete is overexcited or underexcited (e.g., too amped up or too flat) or when an athlete loses control of his or her emotions and ceases to function effectively in a performance environment (e.g., an athlete who cannot control his or her anger or frustration). Training athletes to tap into their emotions to generate or elevate energy, while maintaining a sense of control over those emotions, is a key toward generating an ideal performance state (64).

By arming athletes with the mental tools to combat inappropriate thoughts, enhance confidence, and reinforce motivation and commitment, coaches are providing many of the skills necessary to allow the athlete to maintain composure as well.

Arousal

The training environment provides a host of new and unfamiliar experiences that create multiple opportunities to test oneself and be evaluated for effectiveness. Because of this, athletic performance is frequently affected by arousal, anxiety, and stress. These terms are often used interchangeably; however, in reality, they are different elements within the same construct.

Arousal is simply a blend of physiological and psychological activation in an individual and refers to the intensity of motivation at any given moment (64). For example, a psyched-up athlete may experience tremendous mental activation characterized by positive thoughts and a strong sense of control, whereas a flat athlete could experience minimal activation characterized by wandering thoughts and feelings of boredom. Arousal is always present in an individual to some degree, on a continuum ranging from being deeply asleep, or comatose, to highly excited; but in and of itself it is not automatically associated with pleasant or unpleasant events. It is simply a measurement of activation, and as such can be indexed by such metrics as heart rate, blood pressure, electroencephalography (EEG), electromyography (EMG), and catecholamine levels, or with self-report instruments such as the activation–deactivation checklist (63). The optimal arousal required for efficient performance depends on several factors that are discussed later in the chapter.

Anxiety

Anxiety is a subcategory of arousal in that it is a negatively perceived emotional state characterized by nervousness, worry, apprehension, or fear and is associated with a physiological activation of the body (64). Because anxiety requires the individual's negative perception of a situation, it incorporates a cognitive component, called **cognitive anxiety**, as well as the physical reaction, or **somatic anxiety**, evidenced through physical symptoms such as tense muscles, tachycardia (fast heart rate), and upset stomach.

The term *anxiety* is often confusingly used to refer to both a stable, enduring personality construct and a more short-term, changeable mood state. These are, in fact, different constructs within the emotional state of anxiety. **State anxiety** refers to a subjective experience of apprehension and uncertainty accompanied by elevated autonomic and voluntary neural outflow and increased endocrine activity (57). State anxiety is a generally negative experience, but its effects on athletic performance can be positive, negative, or indifferent, depending on such factors as the athlete's skill level and personality and the complexity of the task to be performed.

State anxiety is distinct from but related to **trait anxiety**, a personality variable or disposition relating to the probability that one will perceive an environment as threatening. In essence, trait anxiety acts as a primer for the athlete to experience state anxiety (58). Trait anxiety also affects the appropriate level of arousal for a given individual. People with high levels of trait anxiety tend to flood attentional capacity with task-irrelevant cognitions, such as thoughts of failure, catastrophe, or ego-oriented concerns. During a complex decision-making task, these attentionally demanding cues could compromise an athlete's selective attention. The athlete with low trait anxiety may handle higher levels of pressure with decreased thoughts of task failure.

State anxiety is the actual experience of apprehension and uncontrolled arousal. Trait anxiety is a personality characteristic, which represents a latent disposition to perceive situations as threatening.

In a non-anxious state, arousal is under the control of the athlete; it can be elevated or lowered as needed. The athlete who is psychologically well prepared knows the appropriate zone for optimal performance and can manage it accordingly. In an over-anxious state, arousal is relatively uncontrolled; muscles are tense, the heart

is racing, and negative thoughts intrude. This lack of physical and **psychological efficiency** is typically initiated by uncertainty about a present or anticipated event. At least three important factors are usually present:

- A high degree of ego involvement, in which the athlete may perceive a threat to self-esteem
- A perceived discrepancy between one's ability and the demands for athletic success
- A fear of the consequences of failure (such as a loss of approval from teammates, coach, family, or peers)

Because these constructs of anxiety and arousal are complex and interrelated, [figure 9.1](#) provides a summary of the interrelationships of arousal, state anxiety, and trait anxiety and the cognitive and somatic components of anxiety.

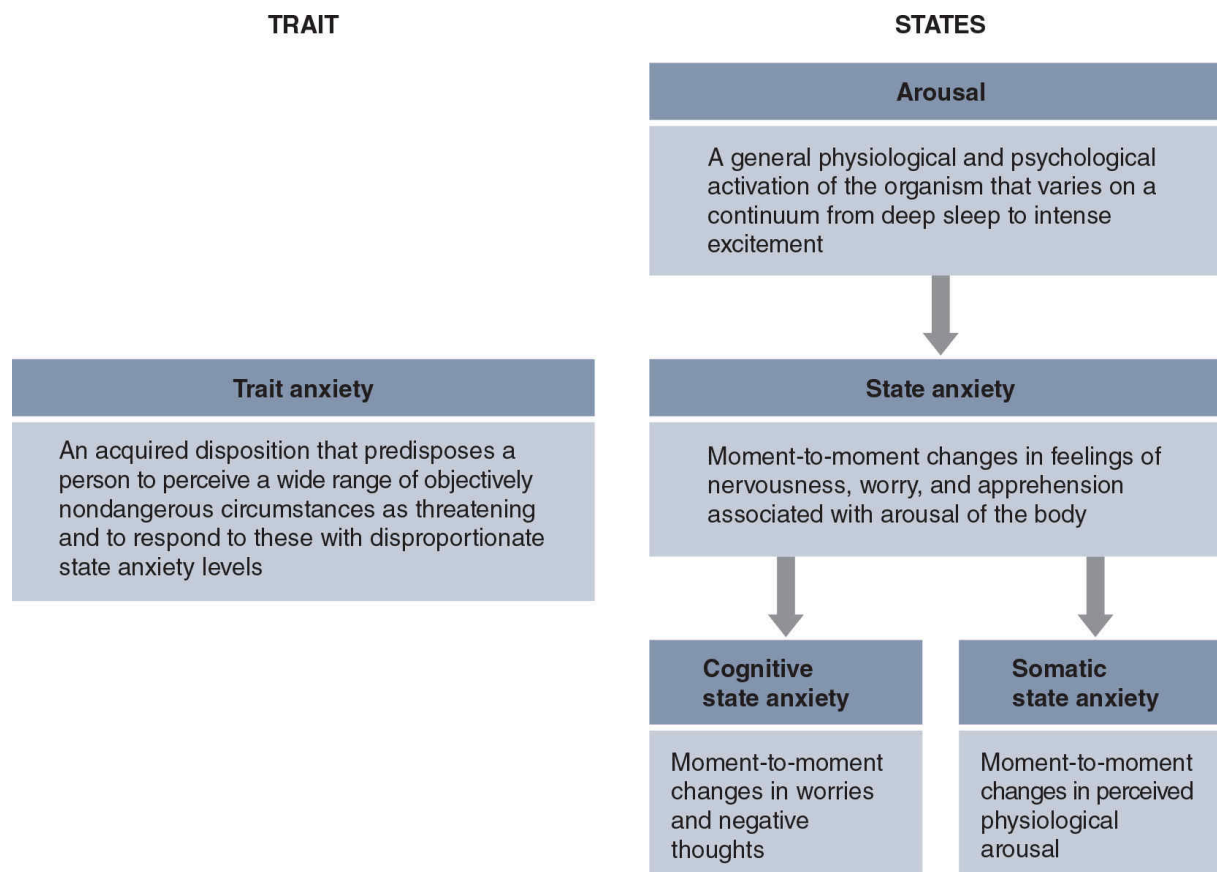


FIGURE 9.1 The interrelationships of arousal, trait anxiety, and state anxiety.

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Stress

Stress is defined as a substantial imbalance between demand (physical, psychological, or both) and perceived response capability, under conditions in which failure to meet that demand has important consequences (51). A **stressor** is an environmental or cognitive event that precipitates stress (i.e., the stress response). Stress can be described as a negative (**distress**) or a positive (**eustress**) state. Both of these generate arousal but only when the perception of the stressor is negative—distress—does it also generate anxiety. Therefore, distress comprises cognitive and somatic anxiety, whereas eustress comprises positive mental energy and physiological arousal.

Theoretical Tenets of Arousal, Anxiety, and Motivation on Performance

Once an understanding of the general concepts of arousal, anxiety, and stress is attained, the next step lies in deconstructing how these emotional elements influence performance. Why does arousal affect one athlete beneficially and another detrimentally? This section examines a number of theories and models that attempt to explain this relationship between arousal and performance. It begins with the simplest construct—Hull’s (56) drive theory—and builds on it with Yerkes and Dodson’s (76) inverted-U theory, describing the mediating influences of skill level, task complexity, and personality. Other related theories that further explain this relationship are outlined, including Hanin’s (23) individual zones of optimal functioning (IZOF), Fazey and Hardy’s (24) catastrophe theory, and finally, Kerr’s (30) reversal theory.

Drive Theory

When researchers first started examining the relationship between arousal and performance, it was thought to follow a direct and linear progression. Hull’s **drive theory** (56) proposes that as an individual’s arousal or state anxiety increases, so too does performance. Thus, the more psyched up athletes become, the better they perform. This may hold true as an athlete progresses from relatively low levels of arousal to somewhat higher amounts, but most would recognize that more arousal is not always better, because performers can clearly be too pumped up to perform well. Depending on the skill and experience levels of the performer, the complexity of the activity, or both, more arousal may be beneficial, but it can in fact also be detrimental. When people perform well-learned or simple skills, a higher level of arousal can benefit performance. However, the more complex a given skill becomes, or the less experience an athlete has with that skill, the more arousal

can produce catastrophic performance outcomes (64). Also, self-confidence serves as a buffer for anxiety, and when self-confidence goes down, anxiety typically increases.

Skill Level

An athlete's skill level can increase the latitude of optimal arousal; that is, the more skill an athlete has developed, the better he or she can perform during states of less-than or greater-than-optimal arousal (64). In the beginning stages of learning a skill, the athlete is in a stage of analysis or cognition (17). This means that he or she has to think about actions. For example, a novice basketball player has to be conscious of the ball while dribbling and needs to devote some attention to the task. At a given level of arousal, worrisome thoughts compete with an attentional capacity already filled with details of motor performance (i.e., dribbling). If a new situation suddenly develops, the novice's mind is already occupied, and he or she may not see it.

The optimal arousal point is lower for less skilled athletes than for more advanced athletes. Therefore, coaches should lower arousal and decrease the decision-making responsibilities of developing or unseasoned athletes (players who are skilled but lack competitive experience) and have them focus on simple assignments to prevent attentional overload. In coaching weightlifting athletes during an important meet, instructions to novice competitors should be simple, clear, and direct. When they experience success, the derived self-confidence may reduce negative self-talk and the sense of uncertainty that typically characterizes such performers.

Task Complexity

A second factor that influences the appropriate level of arousal to achieve optimal performance is task complexity (42). Most athletic skills are exceedingly complex from a biomechanical perspective, but the more concerning complexity relates to conscious decision making. For example, running is a very complex task in terms of

motor control and functional anatomy, but athletes fortunately do not have to devote much conscious attention to this coordinated action. In fact, the action can become altered and inefficient if they think about it too much, because they change the neural sequences for movement initiation. From an attentional perspective, simple or well-learned skills are less affected by a high degree of arousal because they have few task-relevant cues to monitor (64). Fortunately, then, physiological arousal, which typically accompanies emotional arousal, may be beneficial. However, the situation is dramatically reversed for skills that necessitate conscious decision-making effort, such as those required by a goalie in soccer or a baseball catcher who is facing a critical, bases-loaded pitch. In these instances, arousal must be optimized because of the need to maintain a wider focus so as to recognize relevant attentional cues.

Inverted-U Theory

Building on the basic relationship outlined in drive theory, Yerkes and Dodson (76) proposed one of the major tenets of the arousal–performance relationship—the **inverted-U theory**. Basically, this theory states that arousal facilitates performance up to an optimal level, beyond which further increases in arousal are associated with reduced performance. [Figure 9.2](#) graphically shows this relationship. Most coaches and athletes intuitively accept this hypothesis, because they have all experienced poor performance when feeling flat or under-aroused, as well as when they feel too amped up or out of control. The inverted-U concept helps coaches and athletes understand why arousal affects performance and enables them to gain greater control over the appropriate level of arousal for a given athlete within a given sport. It should be noted, however, that the generic shape of the curve in this relationship has been criticized, given our understanding of the individual influences of skill, ability, experience, and task complexity as described previously (20).

Individual Zones of Optimal Functioning Theory

Hanin (23) noted the interactions of individual factors that affect optimal arousal for performance and developed the **individual zones of optimal functioning** theory. Dr. Hanin holds that different people, in different types of performances, perform best with very different levels of arousal. This theory differs from the inverted-U hypothesis in two ways: (1) Ideal performance does not seem to always occur at the midpoint of the arousal continuum, and (2) rather than there being a single defined arousal point at which optimal performance occurs across sporting activities, the best performance can occur within a small range, or bandwidth, of arousal level. Further, Hanin proposed that there are positive and negative emotions (e.g., excited and nervous) that can generate enhanced performance, just as there are positive and negative emotions (e.g., comfortable and annoyed) that can debilitate performance. This proposal is important because it recognizes that any specific emotion can be positively perceived by one athlete but negatively perceived by another. In practice, then, athletes can retrospectively recall the arousal associated with several of their performances that differ in quality. They can then monitor emotions and arousal levels before an important match and make adjustments to increase the chances of falling into this individual ideal zone.

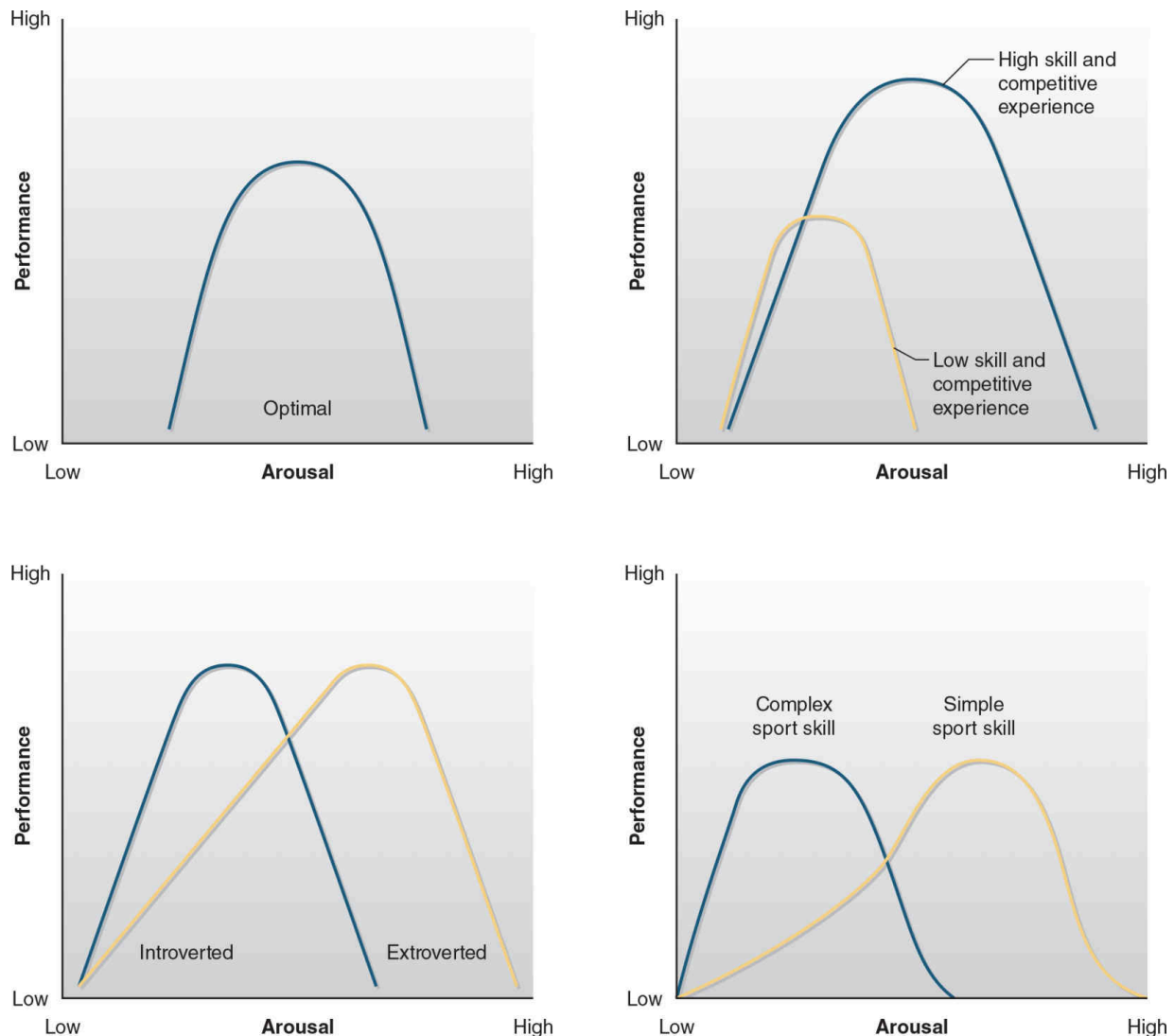


FIGURE 9.2 Inverted-U theory and its modifications.

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Catastrophe Theory

According to Hardy (24), assessment of the cognitive and somatic dimensions of arousal can sharpen the ability to predict (and therefore control) their impact on performance. Earlier assumptions associated with the inverted-U theory held that increases in arousal beyond the optimal level resulted in gradual, proportionate declines in performance. However, common observation tells us that this is not always the case—an athlete may suffer a severe and catastrophic decline rather than a gradual quadratic or curvilinear

decline in performance, and restoring a degree of calm does not necessarily bring a return to the level of performance exhibited before the decline. In this model, the **catastrophe theory**, somatic arousal has a curvilinear, inverted-U relationship to athletic performance, whereas cognitive anxiety shows a steady negative relationship to performance. When increases in physiological arousal occur in the presence of cognitive anxiety, a sudden drop—rather than a gradual decline—in performance occurs. The practical implication of this theory is that the arousal constructs need to be more clearly delineated as cognitive anxiety, physiological arousal, somatic anxiety, or some combination of these.

Reversal Theory

Kerr's (30) interpretation of **reversal theory** posits that the way in which arousal and anxiety affect performance depends on the individual's interpretation of that arousal. Essentially, one athlete might interpret high levels of arousal as excitement and indicative of performance **readiness**, while another athlete, experiencing the same emotion at the same arousal level, would interpret that feeling as unpleasant and reflective of a lack of confidence. This idea implies that athletes have within their power the ability to reverse their interpretation of their own arousal; instead of perceiving high arousal as scary and worrisome, they can choose to reverse their perception and interpret the arousal as reflecting excitement and anticipation. This theory is important because it emphasizes that one's interpretation of arousal, and not just its amount, is significant. Further, it shows that the way in which arousal and anxiety influence performance—whether they are beneficial or detrimental to performance—is within the control of the individual.

Motivation

As mentioned earlier in this chapter, motivation is a primary psychological factor in the acquisition and effective performance of motor skills. **Motivation** can be defined as the intensity and direction of effort (64). Selected aspects of motivational phenomena are highlighted in the following sections. We first discuss intrinsic and extrinsic motivation, which greatly influence the athlete's desire to train and compete. Then we discuss achievement motivation, which helps to explain individual differences in competitiveness. Finally, positive and negative reinforcement are explained as they apply to skill learning and performance.

Intrinsic and Extrinsic Motivation

Intrinsic motivation is important for any athlete. Deci (13) defined this construct as a desire to be competent and self-determining. With intrinsic motivation, athletes are driven because of their love of the game and the inherent reward they feel from participation. This is motivation that comes from within the athlete and is exhibited regardless of the existence of material reward or punishment. Intrinsically motivated athletes focus on the enjoyment or fun they experience in the activity and generally desire to learn and improve because of this love for the behavior in and of itself (64). How can such a desirable state be maintained or encouraged? The answer lies in Deci's (13) definition, which stresses success (competence) and "pulling one's own strings" (self-determination). Appropriate goals, especially for training (i.e., process) or performance (i.e., product), can increase perceived competence. Additionally, giving the athlete some latitude in decision making increases perceived self-determination (13). Although authoritarian behavior is sometimes warranted in sport, in that clear directives are needed in a stressful and competitive environment, a total lack of delegated responsibilities could result in a loss of initiative and drive in athletes (64).

In opposition, **extrinsic motivation** is motivation that comes from some external as opposed to internal source. Numerous examples of this exist, because it is founded on an individualized reward construct; examples of more common extrinsic rewards in sport settings include awards, trophies, praise from coaches and teammates, social approval, and fear of punishment. It should be noted that athletes are rarely completely intrinsically or extrinsically motivated; rather, they exhibit varying degrees along this motivation continuum depending on the activity, their perceptions of competence, the level of importance the activity has, and a host of other variables (11, 13, 38, 64, 68).

Achievement Motivation

Within the general construct of motivation lies a more specifically targeted type called **achievement motivation**, which refers to a person's efforts to master a task, achieve excellence, overcome obstacles, and engage in competition or social comparison. All things being equal between two athletes, whoever is higher in achievement motivation will be the better athlete because he or she has a greater appetite for competition (64).

McClelland and colleagues (38) theorized that all people have opposing personality traits within themselves: the **motive to achieve success (MAS)** and the **motive to avoid failure (MAF)**. The MAS relates to the capacity to experience pride in one's accomplishments and is characterized by a desire to challenge oneself and evaluate one's abilities. On the other hand, the MAF relates to the desire to protect one's ego and self-esteem. Despite its name, however, MAF is not really about avoiding failure itself. It is more about avoiding the perception of shame that accompanies the failure.

Generally, MAS-dominated athletes are most intrigued by situations that are either uncertain or challenging, with an approximate 50% probability of success (64). This creates opportunities to assess one's abilities. On the other hand, MAF-

dominated players are more comfortable in situations in which it is either very easy to achieve success (thereby avoiding shame) or so extremely difficult that there would be no expectation of winning (again, eliminating the likelihood of feeling shame) (64). At higher levels of sport involvement, it is unlikely that athletes would be dominated by MAF, but they would certainly show degrees or a range of competitiveness. Confronted by a very challenging goal, such as gaining a significant amount of lean muscle weight during the hypertrophy phase of a periodized cycle, the MAF-dominated individual might reduce effort because he or she fears failure and the threat to self-esteem (and might also claim that the goal is unrealistic), whereas the MAS-dominated individual might heighten effort in response to the challenge and not perceive any threat.

Motivational Aspects of Skill Learning (Self-Controlled Practice)

In addition to providing the individual with information for skill acquisition, practice, instructions, and feedback can also act as motivational factors for enhancing performance. Practice schedules that address the fundamental psychological needs of autonomy, competence, and social relatedness can influence motivation (73). Of interest in the motor learning literature is the role of **self-controlled practice** in enhancing motivation, performance, and skill learning. Self-controlled practice involves the athlete in decisions related to the practice structure, including when to receive feedback or which skill to practice; it also involves simply asking athletes how they believe they are doing (10, 11, 12, 73). This promotes a more active involvement in the practice session and can enhance feelings of competence and autonomy (11, 12). As a result of this heightened motivation, performance and learning improve (11, 12). Engaging the athletes in some decisions related to the practice schedule is an easy yet effective way to allow them autonomy and ownership in reaching their sport performance goals. The concepts

of instructions, feedback, and practice schedules are discussed in greater depth later in this chapter.

Positive and Negative Reinforcement in Coaching

Coaches can also benefit from understanding the concepts of positive and negative reinforcement and positive and negative punishment as they relate to motivation (37). **Positive reinforcement** is the act of increasing the probability of occurrence of a given behavior (a target behavior, such as correct footwork in basketball, is termed an **operant**) by following it with a positive action, object, or event such as praise, decals on the helmet, or prizes and awards. **Negative reinforcement** also increases the probability of occurrence of a given operant, but it is accomplished through the removal of an act, object, or event that is typically aversive. For example, if the team showed great hustle in practice (i.e., the operant is enthusiasm and hustle), the coach could announce that no wind sprints would be required at the session's end. This coaching reinforcement style focuses attention on what the athlete is doing correctly.

Punishment, on the other hand, is designed to decrease the occurrence of a given operant—that is, negative behaviors such as mistakes or a lack of effort. **Positive punishment** is the presentation of an act, object, or event following a behavior that could decrease the behavior's occurrence. An example is reprimanding a player after a mistake or making an athlete do push-ups or sprints after a fumble. **Negative punishment**, or the removal of something valued, could take the form of revoking privileges or playing time, as in benching. Although coaches use a mixture of both reward and punishment, reinforcement (i.e., reward), or a positive approach, is arguably better because it focuses on what athletes should do and what they did right (termed *specific positive feedback*). Reinforcement (both positive and negative) increases task-relevant focus rather than worry focus. A

task-relevant focus facilitates reaction time and decision making. With reinforcement, athletes also build long-term memories of success, self-esteem, self-efficacy, and confidence. Successful experiences more likely color the athlete's view of competition as desirable and as an opportunity to perform. Of course, coaches may punish unwarranted lack of effort, but it seems ineffective to punish athletes for mistakes if they are making the effort to perform correctly.

How Should Positive and Negative Reinforcement Be Applied?

- Coaches should generally subscribe to a reinforcement strategy to assist athletes to focus on what they do correctly.
- Punishment should be used sparingly, because it increases the likelihood that athletes will focus on what they are doing incorrectly.
- Under conditions that promote a narrow focus of attention, positive reinforcement aids a focus on task-relevant cues, while punishment floods attentional capacity with a predominance of task-irrelevant cues.

Attention and Focus

The athlete's ability to focus can be better understood through the construct of attention. **Attention** is defined as the processing of both external and internal cues that come to awareness. A performer's conscious attention is continuously bombarded with a variety of external stimuli and internal thoughts to which it can be directed. The ability to inhibit awareness of some stimuli in order to process others is termed **selective attention**, and it suppresses task-irrelevant cues (e.g., people on the sidelines, planes flying over the stadium) in order to process the task-relevant cues in the limited attentional space. For a baseball pitcher, task-relevant cues might include the hitter's tendencies and the locations of runners on base.

Selective attention, commonly referred to by athletes as their *level of focus*, is the suppression of task-irrelevant stimuli and thoughts.

American football coaches often exploit the strategic potential of selective attention of their opponents by calling a time-out just before a field goal attempt. During the time-out, the opposing athlete might selectively attend more to task-irrelevant thoughts of self-doubt and the potential for failure, rather than the more beneficial thoughts of self-reassurance or a focus on minor instructional cues. Placekickers can deal with this anxiety and attentional challenge by adopting a ritual or a mental checklist, commonly referred to as a **routine**, that consciously directs thoughts to task-relevant and controllable concerns (e.g., breathing, checking the turf, and stretching the hamstrings).

The important underlying principle is that thinking one set of thoughts actively precludes attending to other worrisome thoughts because of the limited capacity of working memory. This human shortcoming can be used to advantage. Before a lifting performance, for example, the athlete might use key phrases to focus on the task-

relevant cues associated with the lift, such as foot placement, back position, point of visual focus, and knee angle during a squat. This strategy can reduce distractions, which often deter optimal effort. Such focusing strategies can promote mental consistency during the preparatory state, which in turn can promote physical consistency—the hallmark of a skilled athlete.

It is also important to note that the ability to focus attention on task-relevant cues and to control distraction is a skill that can be learned and that improves with increased experience. According to Fitts and Posner's (17) classic theory, the athlete progresses through three stages when learning new motor skills. The first stage, termed the *cognitive stage*, is characterized by effortful and conscious regulation of the movement. That is, the athlete has to think about the details of the task. During the second stage, the *associative stage*, the athlete must focus on the task but is less concerned with the details of the movement. Finally, the athlete achieves the stage of *automaticity*, during which the mind is relaxed and the skill is executed automatically without thinking. Assuming proper instruction and coaching, the relaxed mind focuses only on what is relevant to the task at that moment and, at the same time, automatically filters out all irrelevant cues. Attaining automaticity of action and the clarity of thought that often accompanies it is a goal for many athletes.

Attentional Styles

Nideffer (41) formulated an important concept in sport psychology when he theorized that individuals tend to experience shifting categories of attentional styles during performance. These categories are characterized by two dimensions: direction (internal–external) and width (broad–narrow). The first dimension refers to an introspective versus an externally oriented perspective, whereas the second dimension refers to an integrative (expansive) versus a highly selective orientation. These dimensions each occur on overlapping continuums, creating four quadrants of attentional focus: broad external, in which the athlete assesses the situation by

looking at the environment and various elements within it; broad internal, in which the athlete processes information and develops a strategy; narrow internal, in which the athlete mentally rehearses the upcoming action; and narrow external, in which the athlete specifically focuses on one or two external cues to generate action. These constructs and their relationships to each other can be seen in [figure 9.3](#).

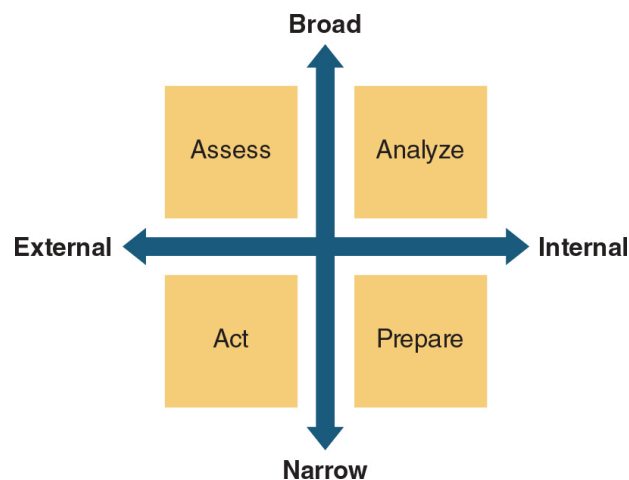


FIGURE 9.3 Four quadrants of attentional focus.

Adapted from Nideffer (1976).

Understanding attentional styles can improve coaching effectiveness. For example, an athlete who tends to become overloaded with external stimuli might be coached to focus on one important cue, such as an opponent's footwork. Athletes who seem to get lost in their own head could practice describing out loud, to a coach, what they are feeling during a lunge. Without such coaching, these players would likely attend to inappropriate cues and react too slowly.

Psychological Techniques for Improved Performance

Improving one's use of mental skills can enhance performance on sporting courts, in training rooms, and on practice fields; it can also transfer to better performance in all areas of life. These psychological skills are quite similar to physical, technical, and tactical skills in that they can be taught and learned and must be practiced on a regular basis if they are to generate long-term behavioral change. This next section introduces several of these psychological skills that can be integrated into training to improve overall performance.

Relaxation Techniques to Control Elevated Arousal and Anxiety

Several techniques can be used to help athletes manage their psychological processes through relaxation. Relaxation techniques are designed to reduce physiological arousal and increase task-relevant focus. These techniques are important when people are executing complex tasks or practicing new ones, when they are dealing with pressure situations, or when deliberate focus is necessary.

Diaphragmatic Breathing

One simple technique for reaching a higher level of physical and mental relaxation is **diaphragmatic breathing** (63). Referred to as *belly breathing*, this form of breathing is a basic stress management technique and a precursor to virtually all other mental training techniques. It focuses attention on the process of breathing to clear the mind and therefore increase concentration. During any mental training exercise, athletes should attempt to engage in deep, rhythmic breathing in a relaxed, natural manner. Physiologically, this form of breathing has a major influence on decreasing heart rate

and muscle tension due to feedback mechanisms that link the respiratory and cardiac control centers in the brain stem. The relatively deep inspiration, followed by a controlled expiration, alters autonomic nervous system (ANS) balance to emphasize parasympathetic neural activity and vagal tone (44). The parasympathetic branch of the ANS promotes the opposite effect of the sympathetic-mediated fight-or-flight response. Thus, rhythmic breathing can decrease neural stimulation of both the skeletal muscles and organs (e.g., heart, lungs, liver), resulting in a sense of deep relaxation.

Diaphragmatic breathing requires that attention be directed to the abdominal region. It is best to get familiar with this starting from a standing position so that breathing is not inhibited. The athlete should let the arms hang loosely and concentrate on relaxing, particularly in the neck and shoulder region, by first taking a couple of deep breaths. Next, the athlete should relax the abdominal muscles so that they appear flaccid. The initiation of each breath should occur simultaneously with the relaxed protrusion of the abdomen; placing a hand on the abdomen gives feedback to ensure that the abdomen protrudes with the initiation of each breath. With each breath the abdomen should become naturally distended. When this portion of the technique is performed properly, the diaphragm (a muscle at the base of the lungs) contracts and drops, allowing a deeper breath to occur. This is the first stage of taking a maximal inhalation. The entire process of inhalation takes place in three different areas and stages: the lower abdomen, the mid-chest, and finally the upper chest. Diaphragmatic breathing can be combined with more dynamic muscular relaxation techniques like progressive muscular relaxation and autogenic training.

Progressive Muscular Relaxation

To achieve an appropriate level of cognitive and somatic activation before performance, athletes may employ progressive muscle relaxation (PMR) (28). **Progressive muscle relaxation** is a

technique by which psychological and physical arousal are self-regulated through the control of skeletal muscle tension. In essence, by going through a series of alternating muscular tensing and relaxing phases, the athlete learns to become aware of somatic tension and thereby control it. The hope is that a relaxed body will promote a relaxed mind.

These tensing and relaxing cycles progress from one muscle group to the next until all muscle groups are optimally relaxed. Each cycle involves maximally tensing each muscle for a short period of time (10-15 s), followed by a conscious attempt to relax that muscle completely, before moving on to the next muscle group. With practice, an athlete learns to rapidly discern the difference between a tense muscle and a relaxed one and then actively take steps to generate the needed muscular relaxation.

In many cases, a positive side effect of the reduced muscle tension is an increase in smooth, fluid, or efficient movement as well as an increased range of motion around the joint. This can be an effective self-regulation technique for some athletes before practice or competition, or even during an intense moment in a given contest. Therefore, athletes should practice this technique in the days leading up to competition (rather than on the day of) to determine its individual impact.

Autogenic Training

For athletes who are injured or who for some reason find it uncomfortable or impractical to experience high muscular tension levels, the PMR cycle for each muscle group can be replaced with an attentional state that simply focuses on the sense of warmth and heaviness for a particular limb or muscle group. This technique, referred to as **autogenic training**, consists of a series of exercises designed to produce physical sensations in the body—generally warmth and heaviness (64). Because autogenic training eliminates the need for uncomfortable levels of muscle tension in the contraction–relaxation cycles, older athletes or athletes in

rehabilitation from injury may find this an attractive alternative to PMR.

Relaxation techniques are designed to reduce physiological arousal and increase task-relevant focus. These techniques are of extreme importance when one is executing complex or novel tasks or performing in high-pressure situations.

Systematic Desensitization

Sometimes fears are learned by association of previously neutral stimuli with a stressful event. For example, an adult non-swimmer who experienced a threatening event in the water as a child may avoid activities around water because of a learned association. This individual may become fearful and therefore tense doing basic resistance or stretching exercises in a pool, even ones that require no swimming skill. This example illustrates the importance of understanding exercise science. For example, the aquatic environment is a great aid in enhancing flexibility. To benefit maximally from a stretching program, however, a participant must learn to fully relax. If the purpose of the pool session is to enhance flexibility, the inability of a non-swimmer to relax in the environment could easily prevent gains in flexibility.

One technique that helps an athlete initially confront or reduce fear is **systematic desensitization (SD)** (71). Systematic desensitization combines mental and physical techniques that allow an athlete to replace a fear response to various cues with a relaxation response. Like cognitive–affective stress management training (SMT) and stress inoculation training (SIT), systematic desensitization teaches the athlete to use a specific relaxation skill-based coping response to control for cognitive arousal (64). This adaptive, learned replacement process, the principle behind SD, is called **counterconditioning**.

To practice the technique, an athlete should be reasonably skilled at both PMR and mental imagery. The athlete should construct a hierarchy, or progression, of events and situations that he or she specifically perceives as fearful. For example, a competitive gymnast who suffered a serious injury on the balance beam may list a series of fearful scenes, proceeding from warming up before the event to the actual movement that precipitated the injury.

In a relaxed setting the athlete visualizes the first scene and experiences a mild degree of anxiety. At the same time, diaphragmatic breathing, PMR, or another physical relaxation technique is instituted, and a strong relaxation response theoretically should overcome the relatively weak **fight-or-flight** syndrome. This technique is practiced until the athlete can hold the image clearly while maintaining a relaxed state. The athlete progresses through the hierarchy, experiencing conditioned fear in small, manageable doses that are overcome by the relaxation achieved with the relaxation technique. This procedure prevents cognitive avoidance and counterconditions a new response (i.e., relaxation) to the formerly fear-inducing stimuli (64).

How Should Athletes Use Arousal Control Techniques?

- An athlete should employ arousal reduction techniques when performing a new skill or one that is complex or when performing in high-pressure situations.
- Athletes should employ arousal enhancement techniques when executing simple skills, ones that are well learned, or in situations of minimal pressure.
- The purpose of employing such techniques is to allow the athlete to perform with an unburdened mind while matching his or her mental and physical intensity to the demands of the task.

Imagery

Imagery is a cognitive skill in which the athlete creates or re-creates an experience in his or her mind (64). Ideally, for athletes, it

uses all the senses to create a mental experience of an athletic performance. The athlete simulates reality by mentally rehearsing a movement, imagining visual, auditory, kinesthetic, olfactory, and even gustatory (taste) cues. Feltz and Landers (16) provided convincing evidence for the effectiveness of mental imagery in the enhancement of sport skill based on a meta-analytic review of the literature. During the initial stages of using imagery, the athlete may start with a relatively simple, somewhat familiar visual image. This helps successful practice of the technique. As with learning any skill by proceeding from simple to complex, the person starts with static images, such as visualizing a golf ball or mentally examining the visual characteristics of a tennis racket. The vividness or detail of the image should become clearer and clearer with continued practice. Some people have a natural talent for achieving image clarity, but everyone can improve with repeated practice.

The perspective of the image can be internal (first person) or external (third person). Although the research literature is unclear as to whether one is superior, it seems that an image that is more engaging and natural to the athlete would be most appropriate. Of course, the internal, first-person perspective seems more specific to skill execution, inasmuch as the actual task is performed with such an orientation. However, because many athletes are familiar with reviewing film or recordings of their performances, an external perspective may also feel natural.

After the athlete successfully visualizes a stationary object with vivid detail, he or she may start to move the object or begin to walk around it in the mind, viewing it from a number of different perspectives. For an image such as a basketball, the athlete may attempt to bounce the ball and feel it against the fingertips. In this manner, the athlete increases complexity by controlling the image or moving it with control (e.g., bouncing the ball) and by bringing a multisensory perspective to bear (i.e., using tactile or kinesthetic as well as visual sensation).

Rehearsing successful execution of a skill during imagined competitive conditions can provide the subconscious mind with positive memories, thus increasing the athlete's sense of confidence and preparedness for the particular sport. Of course, mental imaging is not as powerful a determinant of self-efficacy as actual success, but it does offer two potentially potent ingredients. First, successful performance is entirely under the control of the athlete during imaging, whereas a degree of uncertainty about the outcome is inherent in reality. In imagery the athlete has a great opportunity to experience success. We believe that athletes should be realistic in the kinds of success they imagine; that is, the mental images should be personally challenging yet within the realm of possibility. Second, the athlete can experience competition repeatedly, fostering a sense of familiarity and preparedness.

For some athletes, the months of preparation for a season of play—involving off-season, preseason, and in-season conditioning and skill development—may lead to only a few minutes of actual competitive experience. Even for the starters in team sports, actual competitive experience may be extremely small relative to physical practice time. **Mental imagery**, however, allows the athlete to get used to this uncertain environment over longer periods of time despite minimal real-world competitive opportunity.

Self-Efficacy

Of course, one of the main objectives for applied sport psychology is generating a psychological perspective that improves performance, and it has been argued that perceived self-confidence, or self-efficacy (1), is a better predictor of task execution than either arousal or anxiety. **Self-confidence** is the belief that one can successfully perform a desired behavior (64), while **self-efficacy**, a situationally specific form of self-confidence, is the perception of one's ability to perform a given task in a specific situation (4). Someone who is highly self-efficacious does not doubt his or her ability to succeed at a given task, even when failure is experienced.

According to Bandura's theory (4), a person's self-efficacy is derived from a number of sources:

- *Performance accomplishments*—past experiences of success or failure
- *Vicarious experiences*—watching others (modeling)
- *Verbal persuasion*—encouragement from self or others
- *Imaginal experience*—using imagery to see oneself perform
- *Physiological states*—perception of arousal as facilitative or debilitating
- *Emotional states*—affect or mood

These sources translate well into sport and training environments and can be influenced by the coaches as well as the athlete themselves.

It is thought that if an athlete has the necessary skill set and an acceptable level of motivation, the resulting performance is largely determined by self-efficacy (4). Skill alone is not enough to ensure effective performance—athletes have to want to perform well and need to believe they can be successful in their endeavors. Furthermore, individuals' level of self-efficacy influences the choices made—whether toward certain activities that they feel confident about or away from those about which confidence is lacking. Similarly, this construct additionally influences level of overall effort as well as persistence in the face of obstacles, because those who believe in themselves generally work harder and are more determined to attain the desired outcome than those without the necessary levels of self-efficacy (64). Clearly, the perception of one's ability to successfully perform the tasks of sport or training has a direct impact on the actual performance.

Self-efficacy influences people's choice of activity, their level of effort in that activity, and how much persistence they will have in the face of challenging obstacles.

Self-Talk

A technique frequently used to enhance self-efficacy, aid in directing proper focus, assist in regulating arousal levels, and reinforce motivation is **self-talk**. Self-talk, or intrapersonal communication, is the inner dialogue we have with ourselves (64). It is what we say to ourselves, either out loud or in our heads, that provides the soundtrack for our behaviors and performances.

Self-talk is generally categorized as positive, negative, or instructional and can be either spontaneously generated or more purposefully used to generate changes in mood or behaviors (61). Positive self-talk can include utterances or statements that are encouraging (e.g., "Come on!"), motivational (e.g., "I can do this!"), or reinforcing (e.g., "I am ready!") and generally reflects favorable emotions or feelings. Negative self-talk generally reflects anger, discouragement, doubt, or negative judgment (e.g., "You suck!" "You can't do this." "What were you thinking?"). Instructional self-talk generally provides specific direction or focus on necessary performance cues for a particular skill or strategy (e.g., "Feet shoulder-width apart." "Keep your torso erect.").

Positive and instructionally focused self-talk have been found to improve performance in laboratory settings; however, significant individual and environmental differences may modify these findings in actual performance settings (61). For example, instructional self-talk can be harmful for expert performers, because relaying specific performance cues may actively interfere with the automaticity of the movement. Furthermore, some evidence exists that positive self-talk can result in decreased self-efficacy in some performers (72). Generally, however, negative self-talk is associated with poor performance, because it directs one's focus to inappropriate cues, can trigger negative emotional energy, and can decrease confidence. Therefore, in order to employ self-talk most effectively, athletes should begin by examining their current use of self-talk and its effects on performance. Then, an assessment of the most appropriate way to modify their self-talk can be made.

Goal Setting

Not surprisingly, many of the concepts discussed in this chapter can have a direct influence on one another. Self-efficacy is one of these in that it has a significant impact on the types of goals people set for themselves. Those with higher confidence and efficacy generally envision, create, and strive to accomplish more challenging goals.

Goal setting can be characterized as a process whereby progressively challenging standards of performance are pursued with a defined criterion of task performance that increases the likelihood of perceived success (35). For example, a goal for a swimmer may be to execute a technically correct stroke throughout a sanctioned distance, such as the 50-meter freestyle. In the beginning, the swimmer's skill level might be so low that such a task would seem overwhelming and produce a strong sense of failure and frustration, yet physiological testing may show the coach that the athlete has the physical resources to excel at such an event (i.e., the swimmer has a high degree of fast-twitch, or type II, muscle fibers, exhibits superior muscular power or speed-strength in the upper and lower body, and seems to have a high capacity for anaerobic metabolism). However, the stroke mechanics are inefficient; the athlete is highly aware of this and therefore lacks confidence. First, the coach and athlete can break the skill and conditioning units down into manageable components (the traditional whole-part-whole method of learning). Then as the athlete focuses on and masters each component, a sense of progress and success is nurtured, developing more confidence and better motivating the athlete to complete the challenging goal.

Systematic goal setting can simultaneously increase the psychological development and performance of the athlete (64). A number of reasons explain why goal setting affects performance:

- Goals direct an athlete's attention by prioritizing efforts.
- Goals increase effort because of the contingency of success on goal attainment.

- Goals increase positive reinforcement through the feedback given to athletes.

It seems that the informational nature of thoughtfully derived goals, which increase effort because they are challenging yet attainable, is a powerful ingredient in behavioral change.

Process Goals

An important distinction related to setting goals is the difference between the *process* versus the *outcome* (8). **Process goals** are those over whose achievement the athlete has control. They focus on the actions the individual must engage in during performance to execute the skill well. If the effort is expended, success occurs with a relatively high degree of probability. Examples of process goals within the skill domain relate to form and technique, although an individually determined time (in the case of a swimmer or track athlete) could also be considered a process goal. An example of a process goal within strength and conditioning is to have the athlete focus on the strategy for weight reduction (e.g., what the athlete must do on a daily basis, such as aerobic activity and dietary modifications) rather than on the actual result (e.g., a loss of weight), thereby increasing a sense of control over actions. With process goals, success is strongly contingent on effort.

Outcome Goals

On the other hand, **outcome goals** are ones over which the athlete has little control; typically, winning is the primary focus. Outcome goals in sport generally focus on the competitive result of an event, so earning a medal, scoring points, and generating a high ranking would all fall in this category. The accomplishment of outcome goals is contingent not only on individual effort but also on the efforts and abilities of others—something out of the individual's personal control. We believe that winning is a sound goal orientation that can surely generate high levels of motivation but that, ironically, an athlete can increase the likelihood of its achievement by having both a process

and an outcome goal orientation as opposed to a winning-only attitude. Undue emphasis on winning alone may occupy such a proportion of fixed attentional capacity that it causes narrowing of attentional focus. As such, task-relevant cues are missed, reaction time is slowed, and coordination is diminished by forced movements and compromised automaticity, which alter neuromuscular sequencing—all inhibiting the attainment of the desired goal.

Both process and outcome goals can be applied to strength and conditioning settings as well. For example, emphasizing technique during the power clean illustrates a process orientation, whereas focusing only on completing the set illustrates an outcome orientation. An exception to the general caveat about avoiding outcome goals is the situation in which an athlete is extremely confident and undermatched in competition. He or she may want to focus solely on outcome and a personal best to maximize motivation.

Short-Term Goals

In addition to the process and outcome distinction just discussed, goals can also be categorized as short-term and long-term. **Short-term goals** are generally those that are directly related to current training or competition and are guidelines that can be attained in a relatively short time frame. Short-term goals also increase the likelihood of success because, although challenging, they are relatively close to the athlete's present ability level. They also increase confidence, self-efficacy, and motivation because of the likelihood of success. In this regard, short-term process and outcome goals counteract the boredom and frustration that are potential side effects of long, arduous training regimens.

Long-Term Goals

However, the full meaning of the short-term standards of success is framed by an appropriate long-term goal. **Long-term goals** are those that overarch the series of linked short-term goals. The attainment of those short-term goals should lead to the

accomplishment of the related long-term goal. The athlete may see more relevance in everyday practice goals if it is apparent how they help attain the ultimate level of performance. For example, a gymnast who has a long-term goal of winning the floor routine for the national championship during her senior year may be much more intense and positive about weight room conditioning exercises when she perceives their relevance to her dream. An athlete may be more aroused psychologically and physiologically during practice by the perception that today's activity is another step on the ladder to a personal, long-term dream.

Finally, the specificity of the goals—whether short- or long-term—is important in relation to giving the athlete feedback in effective coaching. Feedback, or the knowledge of success and failure, is more effective in the presence of specific, quantifiable goals, as opposed to vague standards of performance. Feedback is a corrective mechanism, like a thermostat or cybernetic device. Both success and failure can help the athlete stay on course toward long-term success. For example, a specific goal of 25 minutes of continuous running in a heart rate range of 160 to 170 beats/min is a much more engaging objective than “going out for a run.” The vague phrasing may be fine for a recreational participant, but it is not helpful for a competitive cross-country runner, especially when the goal is to develop physiological capacity.

Optimal goal setting requires knowledge of the exercise sciences in both the biophysical and behavioral domains. The efficacy of goals for improving athletic performance lies in their relevance to the physical needs of the athlete. For example, formulating a series of appropriate goals to enable a 400-meter runner to decrease his or her time in the event rests on understanding the physical profile, relevant metabolic pathways, and biomechanical technique to be developed.

Of course, some goals may be completely psychological and therefore only indirectly performance based. An example of such a goal is to adopt a positive mood state for an entire practice.

Although such goals demand less biophysical knowledge, they may be profoundly useful in increasing performance because they are goals that an athlete has tremendous control over and can facilitate inhibiting habitual negative self-talk. However, the most comprehensive goal-setting programs encompass several areas of exercise science in that they might involve goals from any number of subdisciplines, including physiological, metabolic, biomechanical, nutrition, and psychological. This requirement uniquely distinguishes sport psychology from other behavioral sciences.

In general, guidelines for goal setting include the following:

- Long-term goals and short-term goals are interdependent.
- Long-term goals provide a sense of meaningfulness and direction for pursuing short-term goals.
- The attainment of short-term goals provides a hierarchical sense of mastery and success that builds self-confidence.
- Athletes should define process goals to focus on elements of their performance over which they have control.

Yips, Choking, Flow, and Clutch

Often coaches struggle with helping athletes with being less anxious when they compete. Anxiety can interfere with fluid movement patterns. This is known as the *yips*, the *twisties*, or, during critical moments, *choking*. The more technical term is *focal dystonia*.

Simone Biles describes her twisties as a moment when she started thinking and lost air awareness and proprioception as she rotated during a gymnastics competition (58a). "If I still had my air awareness, and I just was having a bad day, I would have continued.... But it was more than that.... It's so dangerous.... It's a miracle I landed on my feet.... As soon as I landed that vault, I went and told my coach that I cannot continue." She expressed a deficit mental focus that created spatial disorientation. Rick Ankiel describes this during his Major League Baseball pitching career like this: "I was pitching with fear ... That put me into a freeze mindset, and that's what led to the yips." Closely related but distinct, choking is when focal dystonia happens during a critical time during performance execution.

Researchers have suggested that perfectionism, social comparison, and dystonia are associated with choking. Opposed to this experience is the positive emotional state of *flow*. Athletes describe flow as effortless movement, complete trust in

movements, and remarkable self-control. Caitlin Clark, a basketball player formerly of the University of Iowa, has said, "It's not about last year. You worry too much about the past, you're going to get caught up in that. It's about being present. Being where your feet are.... Be in this moment." When this happens during critical times during a training session or sport, it is considered clutch. Purposeful focus, optimal emotional control, and high effort describes clutch performances and is often unpleasant during the event. Several stress management techniques can be employed to help work toward flow and clutch, including rhythmic breathing, attentional cuing, and presence-centered focus. However, the yips and choking have been linked to the physiological states of fight-or-flight and, in some case, trauma. *Fight-or-flight* refers to the physiological response of perception of threat that readies the body to defend vigorously or run away. Thus, if athletes are experiencing these issues, professional support is warranted.